

DOCUMENT CONTROL

High-voltage safety and controls

Product-level verification specification for the synthetic Aquila 800 V battery system. Numbered clauses are mandatory unless an explicit logic gate states otherwise.

Table 1. Document control

Field	Value
Document type	Controlled system specification
Authority	Aquila Systems Engineering
Revision / effective date	Rev C / 2026-06-15
Verification evidence	Approved test reports and signed inspection records

Aquila HV safety and monitoring architecture

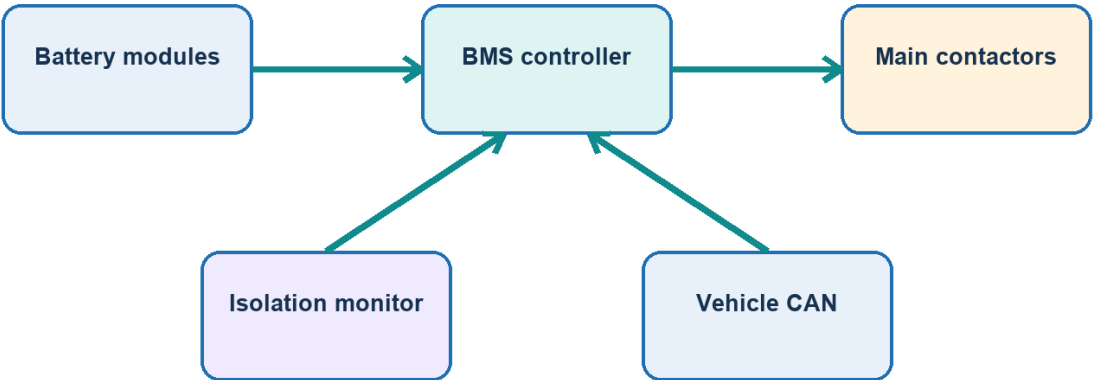


Figure 1. Logical context only; acceptance is defined by the numbered requirements.

Figure 1. System architecture context. The diagram is informative; requirements remain normative.

ELECTRICAL OPERATING ENVELOPE

Electrical operating envelope

MM-REQ-001 - Pack voltage monitoring

The BMS shall continuously monitor pack voltage throughout 400 V to 800 V DC.

MM-REQ-002 - Cell measurement accuracy

Cell-voltage measurement error shall not exceed +/-5 mV from -20 deg C through +60 deg C.

CRASH ELECTRICAL SAFETY

Crash electrical safety

MM-REQ-003 - Crash disconnect response

On a validated crash input, both main contactors shall open within 20 ms and the DC bus shall fall below 60 V within 5.0 s.

MM-REQ-004 - Post-impact isolation

At 60 s after impact, measured isolation resistance divided by the measured DC-bus voltage shall be at least 500 ohm/V.

FLUID INTEGRITY AND THERMAL PROTECTION

Fluid integrity and thermal protection

MM-REQ-005 - Coolant circuit integrity

The coolant circuit shall hold 2.0 bar for 30 minutes with no visible leakage.

MM-REQ-007 - Thermal shutdown and recovery

The BMS shall command shutdown at or above 85 deg C and shall permit recovery only after temperature is below 70 deg C for 60 s.

VISUAL MARKING AND COMMUNICATIONS

Visual marking and communications

MM-REQ-006 - Service-disconnect hazard marking

Every service-disconnect cover shall carry a readable yellow triangular warning label with a black lightning symbol before the cover can be accessed.

MM-REQ-008 - CAN heartbeat endurance

The BMS CAN heartbeat interval shall not exceed 100 ms during a continuous 30-minute run, with zero lost heartbeat messages.

INGRESS PROTECTION AND METROLOGY

Ingress protection and metrology

MM-REQ-009 - Connector J17 ingress protection

Connector J17 shall remain IP67 after 1 m water immersion for 30 minutes, with no moisture inside the rear boot.

MM-REQ-010 - Instrument traceability

Every measuring instrument used for acceptance data shall be identified by serial number and have calibration valid within 12 months on the test date.

CONDITIONAL AND ALTERNATIVE SAFETY LOGIC

Conditional and alternative safety logic

MM-REQ-011 - Drive-away inhibition

If the charge connector is engaged, commanded propulsion torque shall remain zero and vehicle movement shall not exceed 150 mm.

MM-REQ-012 - Alternative post-impact electrical protection

After frontal impact, electrical protection is acceptable when either the isolation ratio is at least 500 ohm/V or an isolation fault causes the main contactors to open within 100 ms.

Logic rule. MM-REQ-011 is IF_THEN. MM-REQ-012 is ANY_OF: proving either alternative is sufficient, but an untested alternative must still remain visible in the audit trace.